

Tanvi Malewar

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Education

Walchand College of Engineering, Sangli

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Bachelor of Technology in Information Technology

9.04/10

Projects

InvigilateX – Automated Invigilation Allocation System

- Developed a full-stack invigilation allocation platform to automate faculty duty assignment using **React.js, Node.js, Express.js, Supabase, and PostgreSQL**.
- Designed and implemented a greedy workload-balancing algorithm to distribute invigilation duties fairly across **200+** faculty members while minimizing workload imbalance.
- Built **JWT-based** role authentication (Admin/Faculty) and integrated CSV parsing for automated bulk import of examination schedules.
- Validated the system using real examination schedules and 200+ faculty records, reducing simulated manual allocation time by approximately 60%.
- Demonstrated the application to the WCE examination coordination team, receiving positive feedback on its feasibility for institutional deployment..

MoneyTrail API

- Developed a backend-only financial transaction service using Node.js, Express.js, MongoDB and Mongoose, following RESTful API principles.
- Implemented JWT-based authentication, user account management, secure transaction processing, and ledger entry generation.
- Designed MongoDB schemas with validation, middleware, aggregation pipelines, and database transactions to maintain data consistency.
- Built features including account balance management, transaction history, idempotency handling, and structured API responses, tested using Postman.

Industrial Sensor Data Analyzer

- Designed an object-oriented C++ system to simulate real-time industrial sensor data monitoring.
- Implemented anomaly detection for identifying out-of-range values and sudden spikes in readings.
- Developed statistical analysis features to compute average, minimum and maximum sensor values.
- Structured application using classes to ensure scalability, maintainability and clean code practices.

Technical Skills

- **Languages:** C++, JavaScript, SQL
- **Frontend:** HTML, CSS, React.js
- **Backend:** Node.js , Express.js , JWT Authentication, REST APIs
- **Databases:** Mongo DB, MySQL
- **Core CS:** Data Structures, Object-Oriented Programming, Computer Networks, Operating Systems, DBMS
- **Developer Tools:** Git, GitHub, Postman, Linux, MongoDB Compass

Achievements

- **2-Star on Codechef**
- **200+ Problems on LeetCode**
- **Runner-up at CloudVerse by SAIT**

Experiences

- Event Manager | Tedx WCE Sangli
- Public Relations Officer | WLUG
- Club Service Director | SAIT
- Technical Speaker at LinuxDiary 6.0 and CloudVerse 2k26
- Organized a club service **Cloud Unbox**
- Conducted a General Interest Meet for about **150+** students